App Dev Project Report

Student details

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About me: I am a student studying data science and machine learning from IITM BS degree online programme.

AI/LLM usage

I used Chatgpt for 15-18%

Description

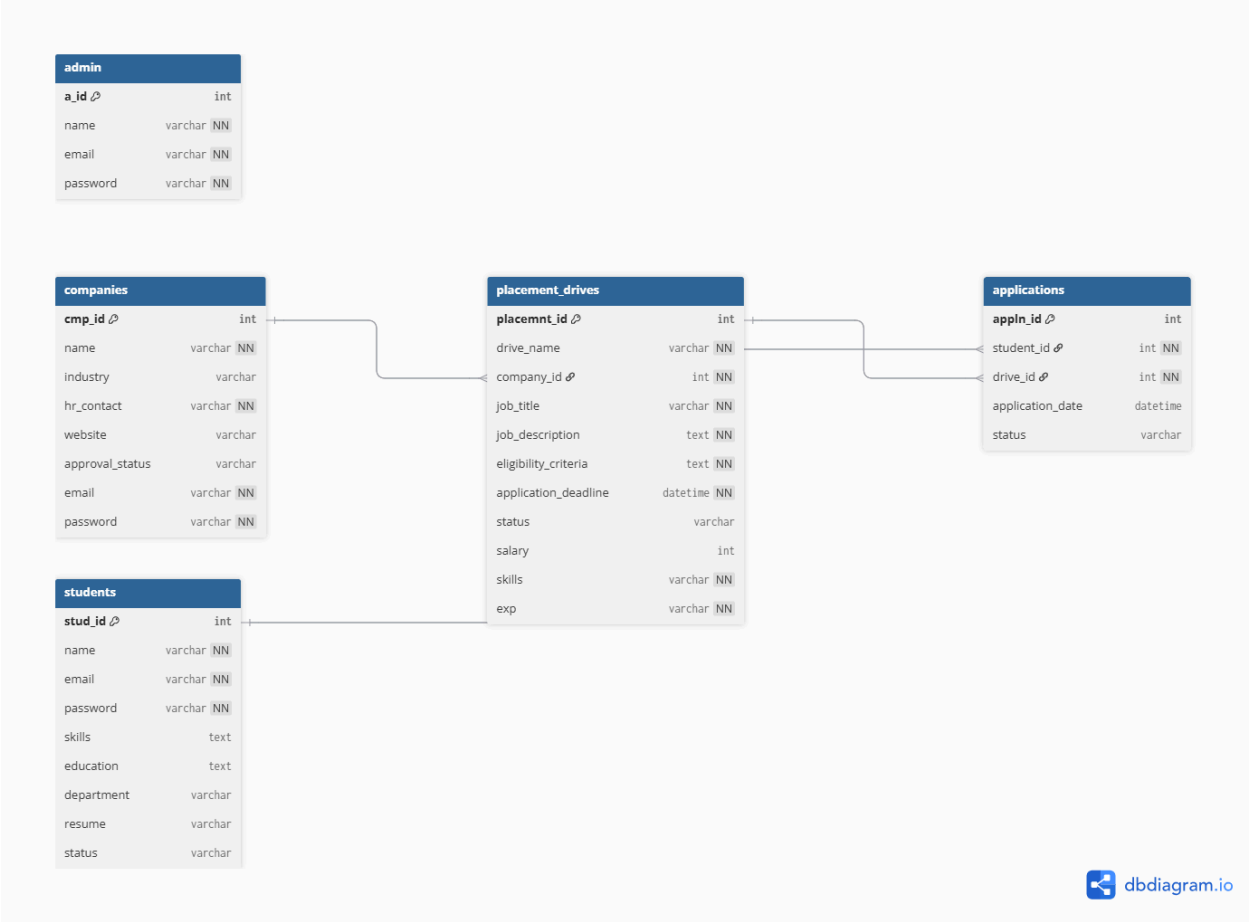
Build a placement portal with Admin, Company, and Student roles using a SQLite-backed Flask app. Admins approve companies and drives, and track applications; companies post jobs/drives and update application status.Students register, apply to approved drives, view history, and receive notifications.

I used ChatGPT for about 13–15% assistance, mainly to draft SQLAlchemy model definitions, correct HTML formatting, and troubleshoot route errors.

Technologies used

Technology	Purpose
Flask	Core backend web framework
SQLAlchemy	Object Relational Mapper for SQLite database
Jinja2	Template engine for rendering dynamic HTML pages
Bootstrap 5	Frontend styling and responsive design
Flask-Login	User authentication and session management
SQLite	Lightweight local database for storing user data
Werkzeug	File upload handling

DB Schema Design



Tables:

Admin — stores administrator account details used to manage the system (a_id, name, email, password)

Companies — stores information about companies participating in placement drives (cmp_id, name, industry, hr_contact, website, approval_status, email, password)

Students — stores student profile and academic information (stud_id, name, email, password, skills, education, department, resume, status)

Placement_Drives — stores details about job opportunities created by companies (placemnt_id, drive_name, company_id, job_title, job_description, eligibility_criteria, application_deadline, status, salary, skills, exp)

Applications — stores student applications for placement drives (appln_id, student_id, drive_id, application_date, status)

Relationships:

One-to-Many → Companies → Placement_Drives

One-to-Many → Students → Applications

One-to-Many → Placement_Drives → Applications

Many-to-Many → Students ↔ Placement_Drives

Architecture and Features

This is a Flask project with the core application setup in `app.py`. The data models live in `backend/models.py` (SQLAlchemy ORM), and all route handlers/controllers are in backend/routes.py. HTML views are stored under `templates/` with role-specific subfolders (`admin/`, `company/`, `students/`), and static assets/uploads are under `static/` (including resumes saved in `static/resumes`). The SQLite database is created programmatically via SQLAlchemy on startup.

Features Implemented

The project implements role-based authentication for Admin/Company/Student, admin approval of company registrations and placement drives, job/drive creation by approved companies, student profile management with resume upload, job search and application, and application status tracking. These are implemented with Flask routes, SQLAlchemy models/relationships, and Jinja2 templates for role-specific dashboards.

Video link:

https://drive.google.com/file/d/1KpQfraQX0MqyfR3ss9akQilnQL1wvi5p/view?usp=sharing